

WILLINGNESS TO PAY FOR IMPROVED RESPIRATORY AND CARDIOVASCULAR HEALTH: A MULTIPLE-FORMAT, STATED-PREFERENCE APPROACH

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SUMMARY

This study uses stated-preference (SP) analysis to measure willingness to pay (WTP) to reduce acute episodes of respiratory and cardiovascular ill health. The SP survey employs a modified version of the health state descriptions used in the Quality of Well Being (QWB) Index. The four health state attributes are symptom, episode duration, activity restrictions and cost. Preferences are elicited using two different SP formats: graded-pair and discrete-choice. The different formats cause subjects to focus on different evaluation strategies. Combining two elicitation formats yields more valid and robust estimates than using only one approach.

Estimates of indirect utility function parameters are obtained using advanced panel econometrics for each format separately and jointly. Socio-economic differences in health preferences are modelled by allowing the marginal utility of money relative to health attributes to vary across respondents. Because the joint model captures the combined preference information provided by both elicitation formats, these model estimates are used to calculate WTP.

The results demonstrate the feasibility of estimating meaningful WTP values for policy-relevant respiratory and cardiac symptoms, even from subjects who never have personally experienced these conditions. Furthermore, because WTP estimates are for individual components of health improvements, estimates can be aggregated in various ways depending upon policy needs. Thus, using generic health attributes facilitates transferring WTP estimates for benefit–cost analysis of a variety of potential health interventions. Copyright © 2000 John Wiley & Sons, Ltd.

KEY WORDS — conjoint analysis; cost–benefit analysis; health valuation; stated preference; willingness to pay

INTRODUCTION

The economic analysis of many health intervention and regulatory programmes often requires evaluating the benefits of improved health. Obtaining credible measures of the economic value of morbidity, however, is one of the more difficult problems facing health economists. The existence of insurance, universal health care systems and

market participants that are unrepresentative of the population of policy interest often obscure essential supply and demand relationships. Thus, revealed-preference information generally has proven to be an insufficient basis for obtaining policy-relevant values of human health. In addition, studies that use contingent valuation (CV) are not well suited to valuing multiple health state attributes. Health economists increasingly are

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turning to stated-preference (SP) approaches as an alternative.^a

This study demonstrates the feasibility of applying SP techniques to elicit values for health conditions described in terms of symptom, activity restriction and duration. Furthermore, by combining two SP elicitation methods, graded-pair and discrete-choice, the estimates presented in this study are more valid and robust measures of benefits than could be obtained from a single format. Our results demonstrate sensitivity to the symptoms being considered as well as to the activity restrictions associated with those symptoms. Subjects clearly indicate systematic preferences for milder morbidity effects over more severe ones. These preferences translate into willingness to pay (WTP) estimates that vary logically with severity.

SP methods evolved as market research tools for evaluating consumer behaviour and predicting sales of new products (Cattin and Wittink [1]; Wittink and Cattin [2]). SP recently has been applied in environmental and health economics as an alternative to CV methods.^b Viscusi *et al.* [3] and Krupnick and Cropper [4] (using the Viscusi data) use SP analysis to elicit a value from subjects for reducing chronic health risks. Other SP studies have elicited preferences for other health and health care attributes, including Hakim and Pathak [5], Vick and Scott [6], Ryan *et al.* [7]; Ryan and Hughes [8], Bryan *et al.* [9], Van der Pol and Cairns [10], Propper [11] and Farrar and Ryan [12].

In this study, specified health states consist of multiple attributes. We presume that people have preferences for different levels within these attributes and are willing to accept some trade-offs among them.^c The utility obtained under different health states is derived from the revealed trade-offs.^d By including a cost attribute, we can use the implicit marginal utility of money to scale changes in health state utility in monetary units.

GRADED-PAIR AND DISCRETE-CHOICE SP QUESTIONS

Two types of SP analysis lend themselves to valuing health effects.^e The first, *graded pairs*, measures subjects' valuations of variations in attributes by requiring them to evaluate trade-offs among various attributes. In a graded-pair for-

mat, subjects sequentially are presented with several different pairs of bundled commodities, represented as sets of attribute levels, and asked to compare each pair. They are asked to rate the intensity of their preference for one of the pairs on a numerical scale, say from 1 to 7, where 1 indicates a strong preference for the first bundle, 7 indicates a strong preference for the second bundle and 4 indicates indifference between the two bundles. The subject is asked to rate a series of these pairs, with each pair having different attributes or attribute levels. (See Viscusi *et al.* [3] for an example of this approach in valuing bronchitis risks.)

Discrete choice, in contrast, confronts subjects with several different products or programmes simultaneously and simply asks them to identify the most-preferred alternative in the choice set.^f As with the graded-pair approach, each commodity is described as a set of attributes. However, the subjects do not provide a rating of the intensity of their preferences. Typically, each subject is shown a series of the choice sets to evaluate.^g

Huber *et al.* [13] argue that each format has relative merits because each relies on a different cognitive process for eliciting preferences.^h The different formats cause subjects to focus on different evaluation strategies. Huber *et al.* [13] cite a well-established result in marketing that the choice format is a better predictor of short-term market behaviour in familiar, repeat-purchase situations, while graded-pairs is a better predictor of longer-term market adaptations to new products with relatively unfamiliar attributes. The health outcomes analysed in this study have attributes of both kinds of situations in different degrees for different subjects. Furthermore, real-world behaviour often involves a mixture of valuation strategies. Thus, we have no *a priori* basis for choosing between graded-pairs or choice-format as the better measure of health preferences in this context. Huber *et al.* [13] have found that employing more than one elicitation format may predict behaviour better than using one format alone, simply because each format requires different heuristics and may provide only a partial picture of preferences. In this study, therefore, we have adopted this approach and apply both graded-pair and discrete-choice formats to estimate joint WTP estimates. In other situations, one or the other approach might be more appropriate.

Table 1 shows the levels associated with each attribute in the two SP formats. We used a simplified Quality of Well Being (QWB) framework for defining health attributes for this study. The QWB health classification system defines health states in four dimensions: three function states (mobility, physical activity and social activity) and the most severe symptom/problem complex. Instead of using the three function states as separate attributes, we combined them into one attribute called 'daily activity'.ⁱ We then added the attributes of duration and annual subject costs to

characterize the desired health state. Pre-tests indicated that higher cost levels were required in the choice format to induce sufficient trade-offs.^j

The experimental design consists of main effects, nearly orthogonal arrays of 40 graded pairs and 40 choice sets using the attribute levels in Table 1. Zwerina *et al.* [14] describe a search procedure for identifying a nearly optimal choice-experiment design. We adapted their algorithm, allowing for the necessity of imposing constraints that eliminate implausible or uninformative combinations of levels and pairs (see Johnson *et al.*

Table 1. Attribute and attribute levels shown in graded-pair and discrete-choice comparisons

Attribute	Level	Description
Symptom	Nose	Stuffy/runny nose and sore throat
	Eye	Eye irritation
	Flutter	Fluttering in chest and feeling light-headed
	Breath	Coughing, wheezing, shortness of breath
	Ache	Coughing or wheezing with fever, chills or aching all over
	Swell	Shortness of breath, and swelling in ankles and feet
	Pain	Pain in chest or arm
Duration	1	5
	1-day episode	5-day episode
		10
Daily activity	NOLIM	You can go to work, go to school, do housework, participate in social or recreational activities and have no physical limitations
	SOMELIM	You can go to work, go to school, do housework and participate in social or recreational activities, but you have some physical limitations (trouble bending, stooping or doing vigorous activities) because of this health condition
	NOSOC	You can go to work, go to school, do housework, but you have some physical limitations (trouble bending, stooping or doing vigorous activities) and cannot participate in social or recreational activities because of this health condition
	ATHOME	You cannot leave your house, go to work, go to school, do housework, participate in social or recreational activities, and you have some physical limitations (trouble bending, stooping or doing vigorous activities) because of this health condition, but you can care for yourself
	NEEDHELP	You cannot leave your house, go to work, go to school, do housework, participate in social or recreational activities, and you need help caring for yourself (feeding, bathing, dressing, toilet) because of this health condition
	INHOSP	You are in hospital and need help caring for yourself (feeding, bathing, dressing, toilet)
Annual costs	Graded-pair (CAN\$)	Discrete-choice (CAN\$)
	10	50
	25	100
	50	200
	100	300
	200	500
	500	750

[15]).^k For example, the relatively mild symptoms 'Stuffy/runny nose and sore throat' (NOSE) and 'Eye irritation' (EYE) are never seen with activity restrictions greater than 'Social and recreation limitations' (NOSOC), nor with costs greater than CAN\$100. Similarly, more severe conditions are never seen with the mildest activity restriction level, NOLIM.^l Given subjects' practical time and attention constraints, we administered eight graded pairs and eight choice sets to each subject by computer, each set of which was drawn randomly from a five-block design of graded pairs and a five-block design of choice sets.

Figure 1 illustrates an example of a graded pair question. In this example, the price is expressed as illness-related costs.^m The subjects indicate their preferences for Condition A versus Condition B. Because there is no unambiguous baseline reference point in the graded-pair questions and no health conditions with zero cost, the graded-pair questions are designed to obtain information on marginal trade-offs among health attributes and costs.

Marketing applications of choice formats generally allow subjects to choose 'current brand' or 'no purchase' (Olsen and Swait [16]; Ruby *et al.* [17]). Including such an opt-out alternative ensures realistic representation of consumer choice, provides an unambiguous reference point from

which to calculate WTP and allows calculation of market shares among competing alternatives. It is not clear what the corresponding opt-out alternative is in a health context. Patients generally do not choose to be sick. Rather, they are endowed with health states and have some control over treatment options. 'No purchase' presumably means remaining in a less-desirable health state for a longer period of time. Returning to normal or good health may require substantial health care expenditures. Thus, we have adapted the market goods choice format to realistically represent health care choices. Current health is the reference point for welfare calculations. Thus, WTP is defined as the monetary equivalent of the utility difference between current health and a specified state of ill health.

Figure 2 illustrates the discrete-choice format used in this study.ⁿ The discrete-choice format directly elicits total values for movements from a given diminished health state to the subject's current health state. The left-hand profile consists of a relatively severe hypothetical initial condition with zero cost. Alternatives A and B represent two courses a subject could choose if experiencing the initial condition. Alternative A portrays a condition of intermediate severity and intermediate cost. Alternative B is described as the subject's current health state on the day of the survey with

Category	Condition A	Condition B
Duration of Episode	5 days	1 day
Symptoms	Eye irritation (itching, burning, redness)	Pain in chest or arm
Daily Activities	• CAN go to work, go to school, do housework • CANNOT participate in social or recreational activities • Have SOME physical limitations • CAN care for yourself	• CANNOT leave your house, go to work, go to school, do housework, and participate in social or recreational activities • Have SOME physical limitations • CAN care for yourself
Total costs of this episode to your household	\$50	\$200

1 2 3 4 5 6 7
 A is A is A is A and B B is B is B is
 much somewhat slightly are about slightly somewhat much
 better better better equal better better better

Please press a number from 1 to 7 that best reflects your rating.

Figure 1. Example of graded-pair question